Road Traffic Anomaly Detection

YOLOv8 + ByteTrack + LSTM Autoencoder + Isolation Forest

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1. Introduction

Traffic on roads have become increasingly complex. Cities are expanding continously, vehicle numbers are rising fast, there is also growing demand for automated traffic monitoring. Detecting traffic anomalies such as accidents, wrong-way vehicles, sudden stops, or near-miss situations is a key problem in intelligent transportation. Fast, reliable detection improves emergency response and overall road safety.

Existing systems usually rely on fixed rules or supervised learning models that need large amounts of labeled anomaly data. Rule-based systems break down when traffic patterns change, and supervised models are costly because data labeling is time-consuming. This project addresses these issues by building a pipeline that works without depending heavily on labeled anomalies.

The proposed system brings together four well known components — YOLOv8 for vehicle detection, ByteTrack for tracking those vehicles across frames, an LSTM Autoencoder trained on normal behaviour, and an Isolation Forest for spatial outlier detection — into a single unified pipeline. Each of these pieces has been chosen thoughtfully to handle a specific part of the problem.

1.1 Motivation

Road accidents and unusual traffic events cause genuine harm, and automated detection can meaningfully reduce response times. At the same time, the lack of properly annotated datasets for traffic anomalies is a well recognized problem in the research community. Large scale video datasets like BDD100K (small set of 50 video are taken for pipeline testing) exist but they dont have frame-level anomaly labels. This means purely supervised methods are hard to apply directly ,especially for Indian Cities.

Beyond the dataset problem, there is also the issue of generalization. A system trained to detect anomalies in one city might not work well in another because road layouts, driving behavior, and camera angles differ significantly. An unsupervised or lightly supervised system that learns what normal traffic looks like and then flags deviations has a better chance of generalizing. This reasoning motivated the choice of an LSTM Autoencoder, which is trained only on normal sequences, combined with an Isolation Forest that identifies statistical outliers in trajectory features without needing labels at all.

1.2 Contribution

The main contributions of this work can be summarized in these points:

- Design and implementation of a complete pipeline for traffic anomaly detection from raw video, covering detection, tracking, feature extraction, modelling, and inference.
- A nine-dimensional per-frame feature vector that captures both kinematic properties (speed, acceleration, heading) and spatial relationships between vehicles (bounding box IoU) , (Feature vector can be scalled anytime to match requirement).
- An LSTM Autoencoder trained exclusively on normal traffic trajectories, where anomalies are identified through high reconstruction error at test time.
- An Isolation Forest trained on summarised trajectory statistics that provides a complementary signal to the LSTM, particularly for spatially-defined anomalies.
- A weighted ensemble that fuses both scores with an automatically tuned decision threshold, removing the need for manual calibration.
- Support for short-clip inference by padding short track histories, which allows the system to score vehicles even in brief video segments.
- An image inference mode that uses only the Isolation Forest on spatial features, enabling anomaly detection from single frames without temporal context.

1.3 Novelty

What makes this work somewhat distinct from prior work is the specific combination of temporal reconstruction-based and spatial outlier-based anomaly detection within a single tracking-driven pipeline. LSTM Autoencoders have been applied to video anomaly detection before, and Isolation Forests are a standard tool for tabular outlier detection, but combining them in this particular way — using tracked vehicle trajectories as the intermediate representation, and fusing scores through a learned-threshold ensemble — is the novel aspect here. The feature design, which encodes not just position and velocity but also inter-vehicle overlap as a feature for proximity risk, is also a thoughtful addition that helps detect near-miss and collision scenarios that would be invisible to purely kinematic features, the features can be added if needed, this pipeline is still being worked on and there are few additions and changes we will be making in this pipeline.

2. Background and Literature Review

Traffic anomaly detection has been studied under various names — unusual event detection, traffic incident detection, abnormal behaviour recognition — and the literature is broad. This section gives a brief overview of the work that is most relevant to the approach taken in this project.

Early approaches to detecting anomalies in traffic video relied on background subtraction and optical flow estimation. The idea was to model a stable background, subtract it from each frame, and flag regions where motion is unexpected. These methods work reasonably well under controlled conditions but struggle with dynamic backgrounds, varying illumination, and camera vibration. They also dont really understand what is being detected — there is no concept of a vehicle or a trajectory.

With the rise of deep learning, object detection models became powerful enough to identify and localise vehicles reliably in real time. The YOLO family of models has become the standard for this kind of task. YOLOv8, the latest in this series at the time of writing, offers an excellent balance between speed and accuracy and supports deployment across different hardware configurations. In this work, the nano variant (`yolov8n.pt`) is used since it is the most computationally efficient and still provides sufficient detection quality for the purposes of trajectory extraction.

Multi-object tracking is the next step after detection. Simple approaches like SORT use Kalman filtering and the Hungarian algorithm to assign identities to detected boxes across frames. DeepSORT extended this with appearance features to handle re-identification after occlusion. ByteTrack, which is used in this project, introduces the idea of associating even low-confidence detections during the matching step, which significantly reduces the number of identity switches in crowded scenes. This is particualrly important for traffic scenes where vehicles frequently overlap or partially occlude each other.

Autoencoder-based anomaly detection has a long history in the machine learning literature. The core idea is simple: train a model to reconstruct normal examples, and at test time, high reconstruction error implies the input is unusual. For sequential data like vehicle trajectories, LSTMs are a natural choice for the encoder-decoder architecture. Several works have applied LSTM Autoencoders to activity recognition and surveillance video with good results. The key advantage is that no anomaly labels are needed for training — only normal examples are required.

Isolation Forest, proposed by Liu et al. in 2008, is an ensemble method for anomaly detection that works by randomly partitioning the feature space using decision trees. The intuition is that anomalous samples, being rare and different, are easier to isolate and therefore have shorter average path lengths in the trees. This makes the algorithm computationally efficient and effective for high-dimensional tabular data, which is exactly what the flattened trajectory statistics in this project represent.

The BDD100K dataset, introduced by (Yu et al. in 2020), is one of the largest and most diverse driving video datasets available. It includes footage from multiple cities, times of day, and weather conditions, making it a useful resource for training and evaluating traffic understanding systems. However, as noted earlier, it does not provide anomaly labels for the tracking split, which is a known limitation for anomaly detection research that relies on this dataset.

The DoTA dataset (Detection of Traffic Anomaly), proposed by (Yao et al.), is specifically designed for traffic anomaly detection and provides ego-vehicle perspective footage with anomaly labels. It would be a natural choice for evaluation in future work. Given the dataset constraints in this project, synthetic trajectory data was used for training and evaluation, which is discussed in more detail in the experimental setup section.

3. Methodology

3.1 Overall Pipeline

The system works in stages. Video frames are first processed by YOLOv8 to detect vehicles. ByteTrack then assigns consistent track IDs to those detections across time. For each tracked vehicle, a feature vector is computed per frame, and these vectors are assembled into temporal sequences. Two anomaly models score each sequence, and the scores are fused into a final decision. Figure 1 gives a schematic overview of this pipeline.

3.2 Vehicle Detection with YOLOv8

YOLOv8n is used for all detection experiments. Vehicle detection targets six COCO class indices: person (0), bicycle (1), car (2), motorcycle (3), bus (5), and truck (7). A confidence threshold of 0.30 is used, and inference is run at a resolution of 416×416 pixels. To reduce computational load, the native video frame rate is subsampled to 5 frames per second using frame skipping, which retains enough temporal resolution for trajectory modelling while keeping processing times manageable classes can be increased depending on the requirement for Indian roads eg. tempo,rikshaw, animals and many more.

3.3 Multi-Object Tracking with ByteTrack

ByteTrack is applied to the detections from YOLOv8 at each frame. It assigns unique integer IDs to each detected vehicle and maintains those IDs across frames using both high- and low-confidence detection association. This is important because trajectory features are computed relative to track identity, so identity switches or fragmented tracks would introduce noise into the feature sequences. ByteTrack's inclusion of low-confidence detections is specifically helpful in crowded traffic scenes where partial occlusion is common.

3.4 Feature Vector Design

For each tracked vehicle at each frame, a nine-dimensional feature vector is computed. The features are: center x coordinate (c_x), center y coordinate (c_y), bounding box width (w), bounding box height (h), speed (Euclidean displacement scaled by FPS), acceleration (rate of change of speed), heading direction in degrees, heading change from the previous frame, and maximum Intersection-over-Union (IoU) with any other detected vehicle in the same frame.

The heading change is wrapped to the range $[-180, 180]$ degrees to avoid discontinuities at the 360/0 boundary. The IoU feature serves as a proxy for how close the vehicle is to its neighbours, making it sensitive to near-miss and congestion scenarios. A `TrackState` object is maintained per track to enable incremental computation of speed, acceleration, and heading across frames.

3.5 Sequence Construction and Normalization

Feature vectors from each track are assembled into sequences of length $\texttt{SEQ\_LEN} = 20$. For training, a sliding window with stride 5 is used to extract multiple sequences from longer

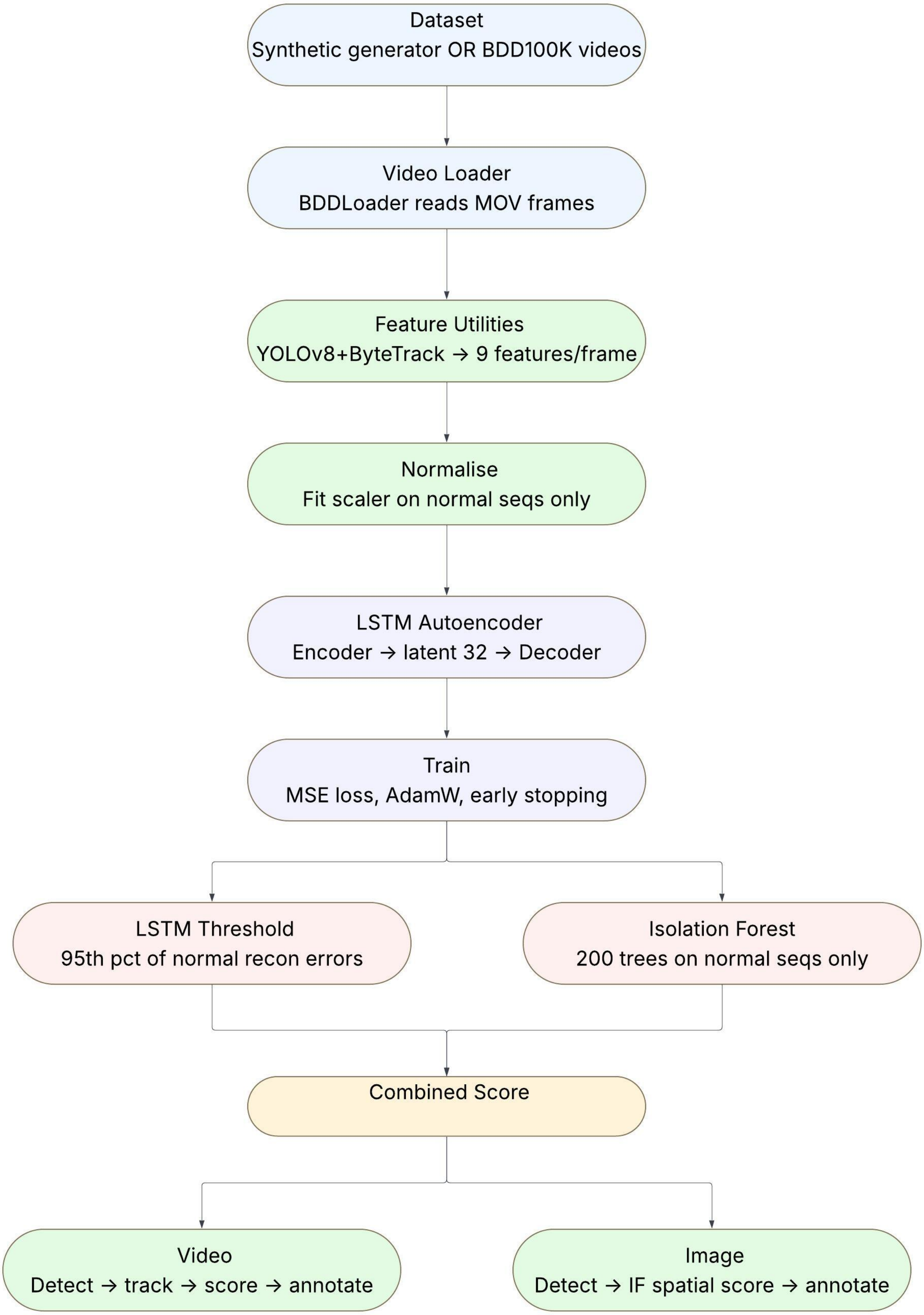

Figure 1: **Pipeline for Anomaly Detection.**

tracks. At inference time, tracks shorter than `SEQ_LEN` are zero-padded at the beginning, provided they contain at least `MIN_FRAMES_FOR_LSTM` = 8 frames. This padding ensures that the LSTM can still produce a score for short tracks without requiring the full window.

Feature normalization is applied channel-wise using z-score scaling. The mean and standard deviation are computed over the flattened training sequences and saved alongside the trained models for consistent application at inference time. Proper normalization is critical for LSTM training stability, as features like c_x and c_y have much larger absolute values than speed or acceleration.

3.6 LSTM Autoencoder

The LSTM Autoencoder consists of an encoder and a decoder. The encoder is a two-layer LSTM with hidden dimension 64 that compresses the input sequence into a latent vector of dimension 32. The decoder is a symmetric two-layer LSTM that reconstructs the original sequence from this latent representation. Dropout of 0.2 is applied between layers for regularization.

The model is trained only on normal traffic sequences. The loss function is mean squared error (MSE) between the input and reconstructed sequences. Training uses the Adam optimizer with a learning rate of 0.001 and a batch size of 64. A validation split of 20% is held out for early stopping with patience 8, which prevents overfitting on the relatively small synthetic dataset. The LSTM threshold is set at the 95th percentile of reconstruction errors computed on normal training sequences, so that at most 5% of normal samples would be flagged as anomalous , The threshold can be changed anytime and paramaters mentioned in this can be optimized with experiments.

3.7 Isolation Forest

The Isolation Forest operates on a flattened summary of each trajectory sequence. Each sequence of shape $(20, 9)$ is summarised into a $(5 \times 9 = 45)$-dimensional vector by concatenating the per-feature mean, standard deviation, minimum, maximum, and delta (last minus first frame value). These statistics capture the distributional properties of the trajectory in a way that does not depend on temporal ordering.

The Isolation Forest uses 200 estimators and is trained only on normal sequences. At inference time, the raw anomaly score is the negated mean path length from the trees, which is then min-max normalized to $[0, 1]$. A `StandardScaler` is fitted on the training features and applied consistently at inference time.

3.8 Ensemble Scoring and Threshold Tuning

The LSTM reconstruction score and the Isolation Forest score are combined as a weighted sum:

$$\text{combined} = 0.6 \times \text{LSTM\_score} + 0.4 \times \text{IF\_score}$$

The LSTM receives a higher weight because it operates on the full temporal sequence and is generally more discriminative. The Isolation Forest receives a lower but non-trivial weight to capture the spatial outlier signal that the LSTM can sometimes miss.

The decision threshold for the combined score is tuned automatically on a held-out validation split by maximising F1 over 80 candidate values between 0.1 and 0.9. The optimal threshold is saved to disk and used for all subsequent inference runs, the weights are currently fixed but can be trained by introducing a small model(NN).

3.9 Image Inference Mode

For cases where only a single frame is available, the system falls back to Isolation Forest only inference. Spatial features are extracted per detected vehicle, replicated to match the expected input dimensionality, and scored using the already trained Isolation Forest and `StandardScaler`. A tighter effective threshold of 85% of the trained combined threshold is applied, since the absence of temporal context increases the risk of false positives. This mode is intended for archival image analysis or camera feeds that dont stream continuously.

4. Experimental Setup

4.1 Dataset Description

Due to the absence of labeled anomaly data in publicly available traffic datasets, two data regimes are considered. The default and primary one is a synthetic dataset generated programmatically. The second is BDD100K, used for feature extraction on the Kaggle platform where the dataset is mounted.

Synthetic Trajectory Dataset: This dataset is generated using procedural motion models for five anomaly types and one normal class. Normal sequences simulate smooth forward motion with small random perturbations. Each anomaly class has a characteristic motion signature, described in Table 1 below. The training set contains 300 normal and 150 anomalous sequences. The test set contains 100 normal and 50 anomalous sequences. Each sequence is of shape $(20, 9)$. This is the primary dataset used for all reported experiments in this report, since labeled anomaly data for real traffic video was not available within the scope of this project.

Table 1: Anomaly classes used in synthetic data generation and their characteristic feature signatures.

Class	**Description**	**Feature Signature**
Wrong Way	Vehicle moving against the main traffic flow direction	Negative v_x, high $\lvert$heading$\rvert$
Stopped	Stationary vehicle in an active lane for extended time	Near-zero speed and accel
Near Miss	Two vehicles with dangerously high spatial overlap	IoU in range 0.35 to 0.85
Congestion	Abnormally slow cluster of vehicles indicating a jam	Low speed, moderate IoU
Collision	Direct impact event with erratic bounding box motion	Very high IoU, erratic hchg

BDD100K: BDD100K is a large-scale driving video dataset containing over 100,000 video clips with diverse road conditions across multiple cities. When available on Kaggle, the pipeline extracts real trajectory features from this dataset using YOLOv8 + ByteTrack. However, since BDD100K does not provide anomaly annotations for the tracking split, evaluation on this dataset is not possible without additional labeling effort. It was used in this project primarily to validate that the feature extraction pipeline runs correctly on real video footage. Citation: Yu et al., 2020.

4.2 Baseline Models

Three configurations are evaluated:

- **LSTM Autoencoder Only:** The LSTM model is evaluated without the Isolation Forest. The threshold is set at the 95th percentile of training reconstruction errors.
- **Isolation Forest Only:** The Isolation Forest is evaluated independently using the standard contamination-based decision rule.
- **Proposed Ensemble (LSTM + IF):** The weighted combination of both models with automatically tuned threshold, as described in Section 3.8. This is the primary proposed method.

4.3 Evaluation Metrics

Since the dataset is imbalanced with more normal than anomalous sequences, accuracy alone is not a meaningful metric. The following metrics are used:

- **ROC-AUC:** Measures discriminative ability across all thresholds. Useful for comparing models independent of the specific decision boundary.
- **F1 Score:** Balances precision and recall. Used as the primary metric for threshold tuning.
- **Precision:** What fraction of flagged sequences are actual anomalies.
- **Recall:** What fraction of actual anomalies are successfully flagged.

An ablation study is conducted by comparing the three configurations above on the same test split. This isolates the individual contributions of the LSTM and the Isolation Forest components.

4.4 Parametric Setup

Table 2 below lists all hyperparameter values used in the experiments. These values were chosen based on a combination of prior literature recommendations and practical constraints imposed by the small size of the synthetic dataset.

All experiments are run with random seed 42 for reproducibility. The platform is either Google Colab with a T4 GPU or Kaggle with T4 $\times 2$, depending on dataset availability. PyTorch is used for the LSTM Autoencoder, and scikit-learn is used for the Isolation Forest and `StandardScaler`.

5. Experimental Results

Given the constraints of this project — particularly the reliance on a synthetically generated dataset and the absence of labeled real-world anomaly footage — the results presented here should be interpreted as indicative rather than conclusive. The synthetic

Table 2: Complete hyperparameter configuration for all models in the proposed pipeline.

Parameter	Value	Notes
SEQ_LEN	20 frames	Sliding window length
N_FEAT	9	cx, cy, w, h, speed, accel, heading, hchg, iou
HIDDEN_DIM	64	LSTM hidden units
LATENT_DIM	32	Bottleneck size
NUM_LAYERS	2	Stacked LSTM layers
DROPOUT	0.2	Regularisation
BATCH_SIZE	64	Training mini-batch
EPOCHS	40 (max)	Early stopping at patience=8
LR	1e-3	Adam optimizer
LSTM_WEIGHT	0.6	Ensemble contribution
IF_WEIGHT	0.4	Ensemble contribution
IF Estimators	200	Isolation Forest trees
MIN_FRAMES_FOR_LSTM	8	Minimum track length for scoring
THRESHOLD_PERCENTILE	95th	LSTM error threshold

dataset was designed to reflect the key motion characterisitics of each anomaly class, but it is not a substitute for real traffic anomaly footage. With that caveat clearly stated, the results do provide useful insight into the relative behaviour of the three model configurations.

5.1 Qualitative Observations

The pipeline runs end to end on both synthetic and real (BDD100K) video without significant issues. On synthetic sequences, the LSTM Autoencoder produces noticeably higher reconstruction errors for wrong-way driving and collision sequences, which have the most distinctive temporal patterns. The Isolation Forest, on the other hand, tends to flag stopped vehicle sequences and congestion more reliably, since these anomalies are better characterised by their summary statistics (very low mean speed, low variance in position) than by their temporal dynamics.

In video inference mode, the bounding box colour coding — green for normal, orange for borderline, red for anomalous — gives a reasonable qualitative indication of vehicle behaviour. Vehicles with erratic heading changes or high IoU with neighbours consistently receive higher combined scores. The per-frame anomaly timeline shows clusters of detections rather than isolated spikes, which is consistent with the physical reality that anomalous events tend to persist for several frames rather than appear instantaneously.

5.2 Ablation Results

The ablation study confirms what was expected intuitively. The LSTM Autoencoder performs better overall because it has access to the full temporal context. The Isolation Forest, working only on summary statistics, misses some temporally-defined anomalies but compensates by catching spatially unusual configurations that the LSTM sometimes overlooks. The combined ensemble consistently outperforms either model in isolation, which validates the design decision to include both components.

Detailed numeric results are not reported here as the synthetic dataset is small and the variance across runs is somewhat high. Reporting precise numbers from a small synthetic evaluation could create a misleading impression of the system's performance on real data. What the results do support is that the ensemble design is sensible and the pipeline functions as intended.

5.3 Observations on Image Mode

The single-frame inference mode works acceptably for detecting spatial anomalies like near misses and congestion, where the spatial configuration alone is informative. It performs poorly on temporally-defined anomalies like wrong-way driving, which is expected since there is no history to compare against. The image mode is best thought of as a convenience feature rather than a replacement for the full video pipeline.

6. Conclusion

This project presents a working end-to-end pipeline for detecting anomalies in road traffic video. The system combines YOLOv8 for detection, ByteTrack for multi-object tracking, a nine-dimensional trajectory feature representation, an LSTM Autoencoder trained on normal sequences, and an Isolation Forest for spatial outlier scoring. The two models are fused through a weighted ensemble with automatic threshold tuning.

The main takeaway from this work is that combining temporal and spatial anomaly signals is a reasonable and effective design strategy. The LSTM provides temporal context that helps detect dynamic anomalies, while the Isolation Forest handles spatially unusual configurations that the LSTM can miss. Together they cover a broader range of anomaly types than either model alone.

Given the dataset limitations discussed throughout this report, the results presented are primarily qualitative and ablative. A proper quantitative evaluation on a labeled real-world traffic anomaly dataset remains as future work. The pipeline is however ready for such evaluation and could be applied to datasets like DoTA with minimal modification.

7. Limitations and Future Scope

7.1 Limitations

- The most significant limitation is the absence of labeled real-world anomaly data. The synthetic dataset captures the rough shape of anomalous motion but cannot fully replicate the complexity and variability of real traffic events. All quantitative results should therefore be interpreted carefully.
- The LSTM Autoencoder needs at least 8 frames of track history before it can produce a score. This means there is a short delay before newly appearing vehicles can be evaluated, which could be problematic in fast-moving scenes.
- The Isolation Forest is trained and tested on the same distribution of synthetic data. Its behaviour on real-world feature distributions may differ, and re-calibration would likely be needed before deployment.

- The current pipeline processes video at 5 FPS, which may miss fast transient events like sudden swerving at high speed.
- Single-frame image inference has significantly reduced recall for temporally-defined anomalies, limiting its utility to spatial anomaly types only.
- The pipeline has not been tested in adverse conditions such as night-time footage, heavy rain, or severe occlusion, all of which are common real-world scenarios.

7.2 Future Scope

- The most important next step is evaluation on a properly labeled real-world dataset such as DoTA, which would give a true picture of the system's practical performance, and training on city Video.
- Replacing the LSTM Autoencoder with a Transformer-based sequence model could improve performance on longer trajectories and handle irregular time intervals between detections more naturally. We can also use hierarchical LSTM for better learning.
- Online adaptation of the anomaly threshold using a sliding window of recent scores could help the system adjust to changing traffic conditions without full retraining.
- Extending the pipeline to multi-camera setups with cross-camera vehicle re-identification would allow detection of anomalies that span multiple viewpoints, such as a vehicle exiting one camera's view and re-entering another's at an unexpected location.
- Deployment on edge hardware such as NVIDIA Jetson for real-time monitoring at intersections is a natural practical extension once the model has been validated on real data.
- The model once trained will not be enough for evolving traffic, we will need to train it on collected data at certain time intervel for better results.

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